山东建筑大学 计算机学院 实验报告

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课程：_移动设备开发实例_____ 同组者：_____ 实验日期：_2024/4/2_____

实验二：Android 移动信息系统初步编程

一、实验目的

1.熟悉 Android 移动信息系统的基本架构。

2.熟练掌握 Android 移动信息系统的基本编程方法。

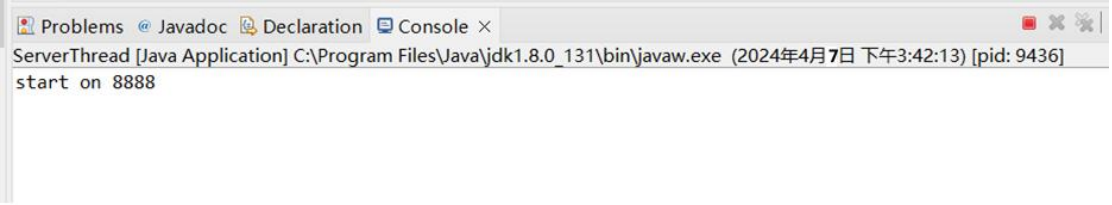
二、实验环境

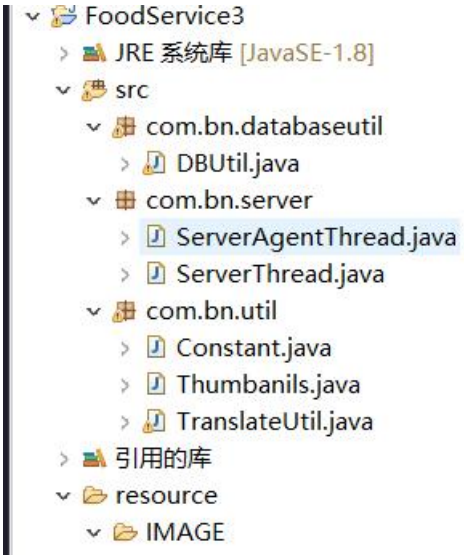
PC 系列微机，Windows。

Android Studio。

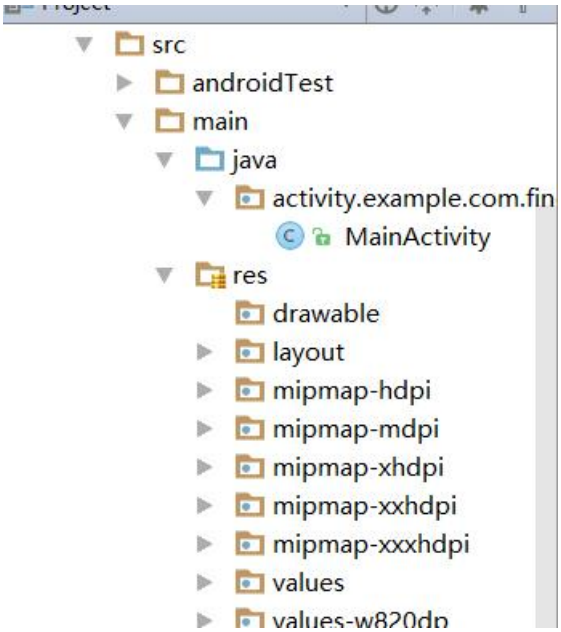
三、源程序

运行" ServerThread.java"文件，控制台显示"start on 8888"信息





前端：



关键代码：

HomeFragmentTest.java

```
public class HomeFragmentTest extends Fragment implements OnClickListener,
        PullToRefreshView.OnHeaderRefreshListener {
    String timediver = "now()"; // 查询时间点
    private Handler mHandler; // //handlrer 更新
    final int DISCON = 0;   //断网
    final int CLOCK = 4; // 定时器
    final int REFRESH = 5;  //刷新
    List<String[]> menuInfo; // 菜谱推荐信息
    ImageDownLoader imageLoader;
    private TextView menu_add;
    private MyGridView menu_ll;
    private PullToRefreshView mPullToRefreshView;
    private FrameLayout fl;
    private ProgressBar pb;
    private LinearLayout again_ll;
    MyToast toast;
    public void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        mHandler = new Handler() {
            @Override
            public void handleMessage(Message msg) {
                super.handleMessage(msg);
                switch (msg.what) {
                    case REFRESH:      //刷新，每刷新一次，都重新加载一下最新信息
                        pb.setVisibility(View.VISIBLE);
                        again_ll.setVisibility(View.GONE);
```

```
                    mPullToRefreshView.onHeaderRefreshComplete();
                    initExcellentMenu();     //在界面上加载精品菜谱
                    fl.setVisibility(View.GONE);
                    break;
                case DISCON:
                    toast.showToast("网络已中断,请联网后重试");
                    fl.setVisibility(View.VISIBLE);
                    pb.setVisibility(View.GONE);
                    again_ll.setVisibility(View.VISIBLE);
                    break;
            }
        }
    };
    new Thread() {
        public void run() {
            while (true) {
                try {
                    sleep(2000);
                    mHandler.sendEmptyMessage(CLOCK);
                } catch (Exception e) {
                    e.printStackTrace();
                }
            }
        }
    }.start();
}
@Override
public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle
savedInstanceState) {
    FrameLayout  home_ll=(FrameLayout) inflater.inflate(R.layout.hometest,null);
    toast=new MyToast(getActivity());
    imageLoader = new ImageDownLoader();
    mPullToRefreshView =
(PullToRefreshView)home_ll.findViewById(R.id.main_pull_refresh_view);
    mPullToRefreshView.lockFooter(); //锁住上拉
    mPullToRefreshView.lockHead();
    //获取精品菜单到menu_ll
    menu_ll = (MyGridView) mPullToRefreshView.findViewById(R.id.menu_ll);
    mPullToRefreshView = (PullToRefreshView)
mPullToRefreshView.findViewById(R.id.main_pull_refresh_view);
    mPullToRefreshView.setOnHeaderRefreshListener(this);
    again_ll=(LinearLayout) home_ll.findViewById(R.id.again_ll);
    pb=(ProgressBar) home_ll.findViewById(R.id.pb);
    fl=(FrameLayout) home_ll.findViewById(R.id.home_nonet);
    Button again=(Button) home_ll.findViewById(R.id.nonet_again);
    again.setOnClickListener(
            new OnClickListener()
```

```
            {
                @Override
                public void onClick(View v) {
                    if (Utils.isNewWork(getActivity())) {
                        new Refresh().start();
                    }else
                    {
                        mHandler.sendEmptyMessage(DISCON);
                    }
                }
            });
        again.performClick();
        return home_ll;
    }
    // /刷新线程
    private class Refresh extends Thread {
        @Override
        public void run() {
            try {
                timediver = "now()";
                menuInfo = NetInfoUtil.getExcellentMenu();     //从数据库中获取精彩菜品
                System.out.println("============1============");
                for(int i=0;i<menuInfo.size();i++){
                    System.out.println(Arrays.toString(menuInfo.get(i)));
                }
                mHandler.sendEmptyMessage(REFRESH);  //发送“刷新”消息，激活 hanlder 继续刷
新
            } catch (Exception e) {
                e.printStackTrace();
            }
        }
    }
    //在界面上加载精品菜谱
    private void initExcellentMenu()
    {
        BaseAdapter ba=new BaseAdapter()
        {
            @Override
            public int getCount() {
                // TODO Auto-generated method stub
                return menuInfo.size();
            }
            @Override
            public Object getItem(int position) {
                // TODO Auto-generated method stub
                return null;
            }
```

```
        @Override
        public long getItemId(int position) {
            // TODO Auto-generated method stub
            return 0;
        }
        @Override
        public View getView(int position, View convertView, ViewGroup parent) {
            String[] strs=menuInfo.get(position);
            String name=strs[1];
            //primaryPic 为菜品图片名
            String primaryPic=strs[2];
            String like=strs[3];
            String collection=strs[4];
            String pinglun=strs[5];
            //先获取 LayoutInflater 对象，再装载列表项布局到布局变量 menu_item
            LinearLayout menu_item=(LinearLayout)
LayoutInflater.from(getActivity()).inflate(R.layout.hometest_menu_item, null);
            int width=Utils.getScreenWidth(getActivity())*7/15;
            //LinearLayout.LayoutParams 用于动态设置控件的布局参数
            //其第一个参数为宽的设置，第二个参数为高的设置。
            LinearLayout.LayoutParams params=new LinearLayout.LayoutParams(width, width);
            ImageView iv=(ImageView) menu_item.findViewById(R.id.image);
            iv.setLayoutParams(params);
            //从缓存或 SD 卡或服务器获取图片并设置到图片控件 iv 上
            imageLoader.thumbnailExcute(iv, primaryPic);
            TextView cname=(TextView) menu_item.findViewById(R.id.name);
            cname.setText(name);
            return menu_item;
        }
    };
    menu_ll.setAdapter(ba);
    menu_ll.setSelector(new ColorDrawable(Color.TRANSPARENT));
    menu_ll.setOnItemClickListener(
            new OnItemClickListener()
            {    @Override
            public void onItemClick(AdapterView<?> arg0, View arg1,int arg2, long arg3)
{
                String[] strs=menuInfo.get(arg2);
                String id=strs[0];
                //Intent intent = new Intent(getActivity(),Menushow.class);
                Intent intent = new Intent(getActivity(), MenushowTest.class);
                intent.putExtra("menu_id", id);
                startActivity(intent);
            }
            });
  }
```

```
    @Override
    public void onClick(View v) {
    }
    @Override
    public void onHeaderRefresh(PullToRefreshView view) {
        mPullToRefreshView.postDelayed(new Runnable() {
            @Override
            public void run() {
                if (Utils.isNewWork(getActivity())) {
                    new Refresh().start();
                }else
                {
                    mHandler.sendEmptyMessage(DISCON);
                }
            }
        }, 1000);
    }
    @Override
    public void onDestroy() {
        // TODO Auto-generated method stub
        super.onDestroy();
        imageLoader.cancelTask();
    }
}
```

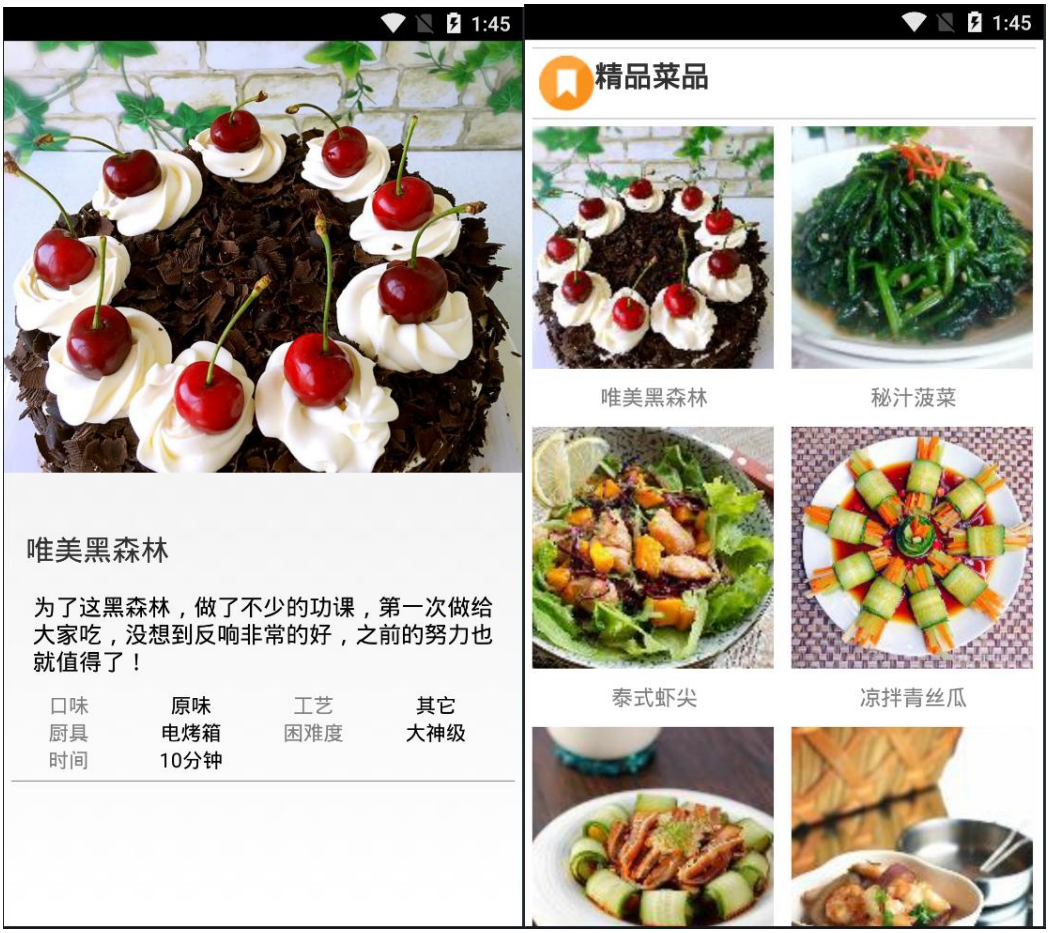
MenushowTest.java

```
/*
 * 菜谱详细信息界面
 */
public class MenushowTest extends Activity  {
    String menu_id;
    Handler mHandler;
    String uid;
    private TextView tv_desc_long;
    ImageDownLoader imageDownLoader;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        // TODO Auto-generated method stub
        super.onCreate(savedInstanceState);
        setContentView(R.layout.menushowtest);
        //Bundle 可以理解为 Android 中用来传递数据的一个容器
        Bundle bundle = this.getIntent().getExtras();
        menu_id = bundle.getString("menu_id", null);
        imageDownLoader = new ImageDownLoader();
        mHandler = new Handler() {
```

```
        @Override
        public void handleMessage(Message msg) {
            super.handleMessage(msg);
            Bundle bundle = msg.getData();
            String[] str = bundle.getStringArray("str");
            if ( str != null) {
                if (!str.equals(Constant.NO_MESSAGE)) {
                    //加载菜谱主信息
                    loadReady(str);
                }
            }
        }
    };
    new Thread() {
        @Override
        public void run() {
            try {
                String[] str = null; // 信息数组
                //向服务器请求并返回菜谱主信息
                String mmsg = NetInfoUtil.getMenuDetC(menu_id);
                System.out.println("======================");
                System.out.println(mmsg);
                str = mmsg.split("<#>");
                Bundle bundle = new Bundle();
                bundle.putStringArray("str", str);
                Message msg = new Message();
                msg.setData(bundle);
                //向 handler 发送消息
                mHandler.sendMessage(msg);
            } catch (Exception e) {
                e.printStackTrace();
            }
        }
    }.start();
}

private void loadReady(String[] str) {
    String primary = str[0];
    final String uidP = str[16];
    uid = str[1];
    String[] vice = new String[] { str[4], str[5], str[6], str[7], str[8] };
    String[] food = (str[9] + "%" + str[10]).split("%");
    loadVice(vice);
    tv_desc_long = (TextView) this.findViewById(R.id.tv_desc_long);
    TextView menu_name = (TextView) this.findViewById(R.id.menu_name);
    menu_name.setText(str[2]);
    tv_desc_long.setText(str[3]);
```

```
        ImageView primaryIv = (ImageView) this.findViewById(R.id.primary_iv);
        imageDownLoader.imgExcute(primaryIv, primary);
    }
    // 口味等信息
    private void loadVice(String[] vice) {
        LinearLayout tl = (LinearLayout) this.findViewById(R.id.vice_info);
        String[] name = new String[] { "口味", "工艺", "厨具", "困难度", "时间" };
        for (int i = 0; i < vice.length;) {
            if (vice[i].equals("null")) {
                i++;
                continue;
            }
            LinearLayout ll = (LinearLayout)
LayoutInflater.from(this).inflate(R.layout.text_text2, null);
            TextView name1 = (TextView) ll.findViewById(R.id.name);
            TextView value1 = (TextView) ll.findViewById(R.id.value);
            name1.setText(name[i]);
            value1.setText(vice[i]);
            i++;
            if (i < vice.length) {
                TextView name2 = (TextView) ll.findViewById(R.id.name2);
                TextView value2 = (TextView) ll.findViewById(R.id.value2);
                name2.setText(name[i]);
                value2.setText(vice[i++]);
            }
            tl.addView(ll);
        }
    }
    @Override
    protected void onDestroy() {
        // TODO Auto-generated method stub
        super.onDestroy();
        imageDownLoader.cancelTaskNow();
    }
}
```

四、运行结果


1:45
唯美黑森林
为了这黑森林，做了不少的功课，第一次做给大家吃，没想到反响非常的好，之前的努力也就值得了！
口味 原味 工艺 其它
厨具 电烤箱 困难度 大神级
时间 10分钟
1:45
精品菜品
唯美黑森林
秘汁菠菜
泰式虾尖
凉拌青丝瓜

实验题：

电影购票
你好，李焕英
猫眼评分 9.5
主演：贾玲，张小斐，沈腾
今天63家影院放映491场
阿凡达
猫眼评分 9.5
主演：萨姆·沃辛顿，佐伊·索达尔...
今天61家影院放映531场
人潮汹涌
猫眼评分 9.5
主演：刘德华，肖央，万茜
今天59家影院放映304场
电影
你好，李焕英
主演：贾玲，张小斐，沈腾
猫眼评分 9.5
今天63家影院放映4
简介
2001年的某一天，刚刚考上大学的贾晓玲（贾玲饰）经历了人生中的一次大起大落，一心想要成为母亲骄傲的她却因母亲突遭严重意外，而悲痛万分。在贾晓玲情绪崩溃的状态下，竟意外的回到了1981年，并与年轻的母亲李焕英（张小斐 饰）相遇，二人形影不离，宛如闺蜜。与此同时，也结识了一群天真善良的好朋友。晓玲以为来到了这片“广阔天地”，她可以凭借自己超前的思维，让母亲“大有作为”，但结果却让晓玲感到意外.....
特惠购票

五、实验体会

通过本次实验，加深了对于 Android 开发的理解。熟悉了 Android 移动信息系统的基本架构。掌握 Android 移动信息系统的基本编程方法。